

ABOLFAZL RASTGAR MOGHADAM

Data Engineer | SQL & Pipeline Specialist

ablfzl.rstgr@gmail.com · +98 935 224 7312 · github.com/abolfazlrastgar · Tehran, Iran · Open to relocation

Professional Summary

Data Engineer with 4+ years of production experience building and operating large-scale data platforms in a high-stakes financial/banking environment. Proven track record in end-to-end ETL/ELT pipelines, near-real-time CDC integration (Kafka, Debezium, Spark), and OLTP-to-analytics transformation. Expert-level SQL/T-SQL with strong data modeling (star/snowflake, data marts) and complementary Python for processing and automation. Graduate ML research background (Random Forest, Scikit-Learn) adds depth to data quality, anomaly detection, and observability work.

Core Technical Skills

Programming: T-SQL (Advanced), Python (Pandas, NumPy, Scikit-Learn), SQL, Bash, JavaScript

Databases: SQL Server (Expert — OLTP/OLAP, multi-TB), PostgreSQL (Proficient), ClickHouse (Intermediate), Elasticsearch, Redis

Data Engineering: Apache Kafka, Debezium (CDC), Apache Spark, SSIS, ETL/ELT, Batch & Near-Real-Time Pipelines, Incremental Loads, Change Tracking, Data Reconciliation, SQL Agent

Data Modeling: Star & Snowflake Schema, Data Mart Design, OLTP-to-Analytical Transformation, Covering/Filtered Indexes, Query Optimization, Partitioning

Analytics & BI: Power BI (developing — DAX fundamentals), Metabase, SQL-based Reporting, KPI-driven Data Structures

Observability: Grafana, Prometheus (SQL/Windows exporters), Zabbix, Redgate SQL Monitor, Pipeline Alerting, Freshness & Job Failure Detection

Infrastructure: Docker, Linux (LPIC-1 in progress), Windows Server, AlwaysOn AGs (WSFC), HA/DR, Backup & Encryption

ML / Analytical: Scikit-Learn, Random Forest, Cross-Validation, Feature Engineering, IDS Benchmarks (NSL-KDD, CICIDS2018), MATLAB

Professional Experience

SQL Server DBA & Data Engineer

City Innovation & Development · Tehran, Iran · Financial/Banking Sector Feb 2022 – Present

Data Platform & Pipeline Engineering

- Designed a centralized banking data platform integrating 50+ SQL Server instances (multi-TB) into analytical layers for dashboards and decision-making.
- Implemented near-real-time CDC pipelines (Kafka, Debezium, Spark) moving OLTP changes into a Lakehouse environment with sub-minute latency.
- Built self-service analytics layers (Metabase, Power BI) and high-volume operational analytics layers for business and technical teams.

- Developed SSIS batch ETL pipelines into a reporting DW with transformations, cleansing, incremental loads, and failure alerting; engineered cross-platform reconciliation for account-opening data across heterogeneous banking systems.

Data Modeling & Query Optimization

- Designed star/snowflake schemas and data marts optimized for BI and near-real-time queries.
- Reduced MDF footprint ~90% on a ~10 TB estate and removed ~30% redundant data from a ~5 TB estate, cutting I/O overhead significantly.
- Optimized report latency via covering indexes, filtered indexes, and result caching on multi-TB datasets.

Observability & Monitoring

- Built Grafana/Zabbix dashboards (Prometheus exporters) covering AG health, job status, blocking/deadlocks, CPU/memory/I/O, and storage trends.
- Deployed Redgate SQL Monitor for query diagnostics; established SMS alerting for AG failovers, job failures, space thresholds, and data freshness violations.

High Availability & Infrastructure

- Architected multi-site AlwaysOn AG infrastructure (~30 DBs, ~5 TB, 3-node WSFC across two sites) with DR RTO < 15 minutes.
- Built 3-node WSFC for PardakhtSazi (8 DBs, ~600 GB); tuned OLTP workload for p95 latency improvement. Designed Full/Diff/Log backup strategy saving ~1 TB storage.

Technical Support Engineer & SQL Optimization Specialist

Tirajeh Rayaneh Tehran · Tehran, Iran · Healthcare IT

Mar 2021 – Feb 2022

- Diagnosed and resolved query performance bottlenecks in critical hospital databases (Baqiyatallah Medical Center) using SQL Profiler; index redesigns measurably improved response times.
- Identified application-level data access bugs through Profiler traces; designed zero-downtime migration plans with structured testing and rollback procedures.

Selected Data Engineering Projects

Centralized Banking Data Platform — Real-Time Integration Layer

SQL Server · Kafka · Debezium · Spark · SSIS | 2022 – Present

End-to-end platform ingesting operational changes from 50+ SQL Server instances via CDC streaming and batch ETL. Full observability on multi-TB financial data.

Cross-Platform Account-Opening Reconciliation Service

SQL Server · SSIS · T-SQL | 2022 – 2023

Central reporting service aggregating account-opening data across heterogeneous banking platforms. Standardized ETL schemas, automated reconciliation, and optimized queries for near-real-time delivery.

Data Pipeline Observability & Alerting System

Grafana · Prometheus · Zabbix · T-SQL · Redgate | 2022 – Present

Production observability platform covering DB health, pipeline freshness, job failures, and infrastructure metrics with Prometheus exporters and baseline-aware alerting.

AI-Based Intrusion Detection System — Smart Grids

Python · Scikit-Learn · NSL-KDD · CICIDS2018 | 2022 – 2023

Full ML pipeline (Random Forest, k-fold CV, multi-class classification) for network intrusion detection in smart-grid environments, applicable to production anomaly detection.

Education

M.Sc. Computer Engineering

Sep 2020 – Dec 2022

Iran University of Science and Technology (IUST) · Tehran, Iran

Thesis: AI-Based Intrusion Detection in Smart Grids (Random Forest, Python, Scikit-Learn, NSL-KDD, CICIDS2018)

B.Sc. Computer Engineering

Graduated ~2017

Sadjad University · Mashhad, Iran

Thesis: Shortest Path Search Using Ant Colony Optimization (MATLAB)

Certifications

- SQL Server Database Administrator — Farata az Danesh Academy
- SQL Server Performance Tuning — Farata az Danesh Academy
- Linux LPIC-1 — In Progress
- Apache Kafka, Apache Spark, PostgreSQL, ClickHouse — In Progress (self-directed)

Languages

Persian (Farsi): Native

English: IELTS Academic: ~6.5 Overall (Speaking 6.5)